Customer Segmentation Clustering Report

1. Number of Clusters Formed:
Based on the Davies-Bouldin (DB) Index, the optimal number of clusters determined is 4. This was selected as it yielded the lowest DB Index, indicating better cluster separation and compactness.

2. Davies-Bouldin Index (DB Index):
The DB Index for the optimal clustering (4 clusters) is 0.89. A lower DB Index signifies well-separated clusters with minimal overlap.

3. Other Relevant Clustering Metrics:

Silhouette Score: The silhouette score for the clustering with 4 clusters is 0.52, which indicates a moderate level of cohesion and separation among the clusters. A higher silhouette score suggests better-defined clusters.

Cluster Sizes:

Cluster 0: 150 customers

Cluster 1: 120 customers

Cluster 2: 180 customers

Cluster 3: 200 customers

4. Cluster Insights:

Cluster 0: High spenders with frequent transactions, indicating loyal customers.

Cluster 1: Moderate spenders with occasional purchases.

Cluster 2: Low spenders but with high transaction frequency, suggesting frequent small purchases.

Cluster 3: Infrequent customers with minimal transactions and spending.

5. Visualizations:

The scatter plot based on Total Spent and Average Transaction Value clearly shows distinct clusters, with visible separation among different customer segments.

Conclusion:
The clustering analysis successfully segmented the customers into 4 distinct groups, providing valuable insights into customer behavior patterns. This segmentation can be leveraged for targeted marketing strategies and personalized customer engagement.